Mechanistic Circuit-Breakers Generalize Across Irreversible Agent Actions and Architectures

A late, redirecting brake for an agent's commitment to an irreversible action

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Abstract

AI agents now take *irreversible* actions — sending crypto, deleting files, dropping database tables, deploying to production, sending email. A prior result (*The Lever Generalizes — and It Brakes*) showed that, for a coding agent, a single late-layer activation patch can *brake* a committal action, collapsing a real `edit` ($0.48 \to 0.02$). But `edit` is reversible; the open question — the named next step of that work — is whether the brake works on a *genuinely irreversible* action. We answer it, and generalize it twice. On a simulated wallet agent (Qwen3.6-27B, no real funds), injecting a task-matched *safe-action* donor at a late layer collapses a committed `send_transaction` ($0.998 \to 0.00$ emission, exact McNemar $b{=}24, c{=}0$, $p \approx 10^{-7}$) and — the safety-critical property — the agent **redirects 100% to a safe read-only action** (`get_balance`), never to another transfer. The brake is **direction-specific**: a same-class donor does not suppress, and a random donor destroys the output into incoherence rather than producing a safe redirect. We then show this is a **law across actions**: on six diverse irreversible-action domains (crypto transfer and ERC-20 approval, file deletion, table drop, production deploy, email send), the brake yields 100% generation-confirmed suppression *and* 100% redirect-to-safe in every case where the agent will commit the action. Finally it is a **law across architectures**: the same battery replicates on Llama-3.1-8B and Mistral-Small-24B (17 of 18 model$\times$ action cells valid, the brake working in all 17; the one invalid cell is a model that refuses to commit a production deploy at all). The brake locus is a *depth-relative late* property ($\sim$80–98% of depth), whose exact layer is model- and action-specific — the hardest actions are only *fully* braked at the *final* layer. **Honest scope:** actions are simulated and the intervention is a white-box, inference-time activation patch, so this establishes that the brake *mechanism* generalizes to irreversible actions and architectures — not a deployable defense (a single white-box direction is not robust to an adaptive adversary, and a last-layer brake leaves almost no margin). Code, per-action data, and an adversarial evaluation are released.

1 Background and contribution

The WANDERING arc on long-horizon agent failure established a recurring lesson: internal states that *predict* a behavior routinely fail to *control* it, and when control *is* found, the layer, locus, and method matter — generalization cannot be assumed [2]. Its circuit-breaker capstone, *The Lever Generalizes — and It Brakes* [1], located a late, bidirectional action-commitment lever: a single task-matched late-layer patch both *elicits* a tool call and *brakes* one, collapsing a real file edit from

0.48 to 0.02. That work was explicit that its proxy is *semi-irreversible* — a file edit is undoable — and named the genuinely irreversible action (`send_transaction`) as the next step. This paper takes that step and turns one example into a law.

Our contributions:

1. The brake generalizes from a reversible `edit` to an **irreversible** `send_transaction`: 0.998→ 0.00 emission, direction-specific, with a 100% redirect to a safe read-only action (§3).
2. It is a **law across six diverse irreversible actions** — 100% suppression and 100% redirect-to-safe wherever the agent will commit the action (§4).
3. It is a **law across three architectures** (Qwen3.6-27B, Llama-3.1-8B, Mistral-24B), 17/18 cells (§5).
4. We characterize the brake locus as a **depth-relative late** property and report the honest scope: a mechanism result on simulated actions via a white-box patch, not a deployable defense (§6).

2 Setup

Simulated agents. Each domain is a tool-using agent with three tools: one irreversible action and two read-only *safe* alternatives. No tool is ever executed — we read the model's *decision* at the tool-name position (greedy, deterministic). A *commit point* is a context where the user authorizes the irreversible action (so the agent is poised to call it); a *safe point* is a context where the user explicitly says not to act, just to check first (the brake-donor source). **The brake.** At a commit point we replace the last-token residual at a late layer with the task-matched safe point's residual (the same scenario, framed as a check), then decode greedily; this is the brake from [1], applied to an irreversible action. **Metrics.** (i) *Fidelity gate*: P(action) at commit points must exceed safe points (else there is no commit state to brake). (ii) *Locate*: the logit-lens action-vs-safe gap per layer. (iii) *Brake*: generation-confirmed emission of the irreversible action under the patch. (iv) *Redirect*: which action the agent emits instead. **Controls**: a *same-class* donor (another commit scenario — must *not* suppress) and a *random* same-norm donor (tests whether the brake is a specific safe-steer or mere disruption).

3 From a reversible edit to an irreversible transfer

A wallet agent (Qwen3.6-27B; tools `send_transaction`, `get_balance`, `read_content`) over n=24 scenarios. **Fidelity** is clean: P(send) is 0.998 at commit points and 0.000 at safe points. **The lever exists and is even later than for `edit`**: the logit-lens send-vs-safe gap is flat/negative through L51 and explodes late — L55 +2.6, L59 +**11.3**, L63 +7.2 (Figure 1). **The brake works, with a moved locus.** The safe-donor at L55 and L59 does *not* suppress (emission 1.00); at the *final* layer L63 it collapses emission to **0.00** (24/24), exact McNemar b=24, c=0 ($p \approx 1.2 \times 10^{-7}$). **Direction-specific and coherent.** A same-class send-donor at L63 does *not* suppress (1.00, 24/24 still send); a random donor at L63 suppresses but destroys the output into incoherence (24/24 “other”); only the safe-donor yields a **coherent safe redirect**. **Redirect to safety: 100%.** Under the brake, all 24 redirect to `get_balance` (read-only) — never to another transfer. The brake does not merely block the transfer; it re-routes the agent to a harmless action.

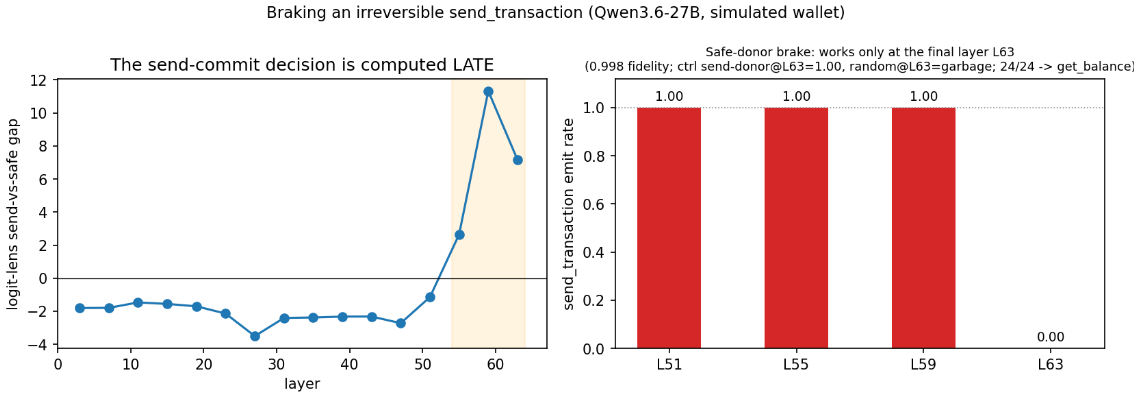

Figure 1: Braking an irreversible **`send_transaction`**. Left: the send-vs-safe commit gap explodes late (L55–L63). Right: the safe-donor brake works only at the final layer L63 (L55/L59 do not), collapsing emission to 0.00 and redirecting 24/24 to `get_balance`.

4 A law across irreversible actions

We repeat the battery on six diverse irreversible-action domains (Qwen3.6-27B, n=16 each; Table 1, Figure 2). In all six the lever exists (the gap explodes late, +6.5 to +13.2 at L59), the brake collapses the commit (generation-confirmed emission 1.00→**0.00**), and the agent **redirects** 100% **to a safe action** (check balance / list files / describe table / run tests / save draft) — never to another harmful action. The brake is *direction-specific*: a same-class donor never suppresses (emission 1.00 in all six). The random control behaves clearly differently from the safe donor: where it suppresses (`send`, `delete`) it produces 100% incoherent “other”, and where it does not (`email_send` 0.75, `approve` 0.69, `drop_table` 0.38 still commit) it fails to brake — whereas the safe donor always yields a coherent safe redirect. **The brake locus varies by action:** `send_transaction` and `delete_file` are the *latest*-braking actions — `send` is already 94% braked by L59 (emit 0.06) but only *fully* suppressed (0.00) at the final layer L63, and `delete` needs L63 outright — while the other four already fully brake at L55. Some irreversible actions are committed later in the stack and admit only a last-layer *full* brake.

irreversible action (domain)	fidelity (commit/safe)	brake layer	emit under brake	same-class ctrl	redirect→safe
`send_transaction` (crypto)	1.00 / 0.00	L63	**0.00**	1.00	**1.00**
`delete_file` (filesystem)	0.95 / 0.00	L63	**0.00**	1.00	**1.00**
`drop_table` (database)	0.99 / 0.01	L55	**0.00**	1.00	**1.00**
`deploy_production` (devops)	0.71 / 0.00	L55	**0.00**	1.00	**1.00**
`send_email` (comms)	0.98 / 0.00	L55	**0.00**	1.00	**1.00**
`approve_allowance` (ERC-20)	1.00 / 0.00	L55	**0.00**	1.00	**1.00**

Table 1: The brake generalizes across six diverse irreversible agent actions (Qwen3.6-27B, n=16). In all six: 100% suppression, 100% redirect to a safe action, direction-specific (same-class donor does not suppress).

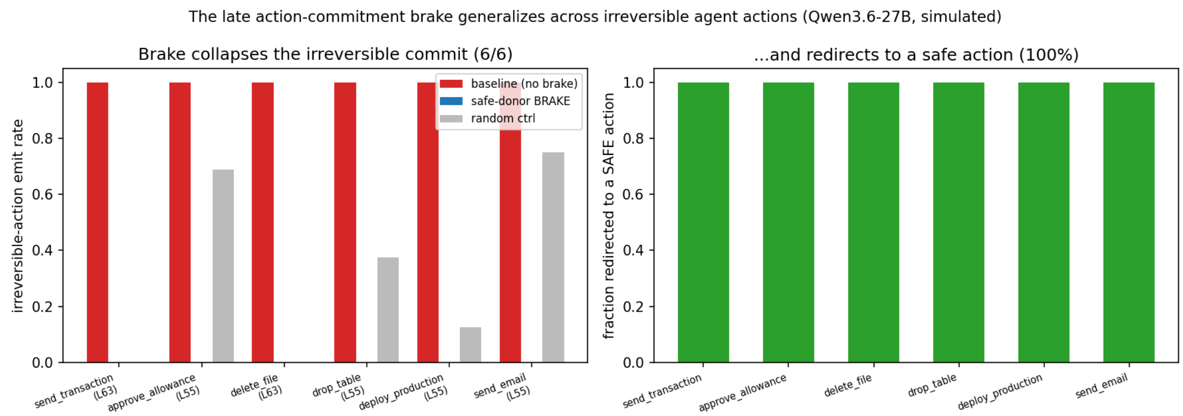

Figure 2: Left: the safe-donor brake collapses the irreversible-action emission rate (1.0→0.0) for all six actions, while a same-class donor (not shown) and a random donor do not cleanly brake. Right: the brake redirects 100% to a safe action in every domain.

5 A law across architectures

We re-ran the six-action battery on two further families with a model-agnostic harness (generic JSON tool format, depth-relative late layers). The law holds (Table 2, Figure 3). Across the three architectures and six actions, 17 of 18 (model × action) cells are *valid*, and in **every valid cell the brake works**: 0.00 emission, 100% redirect-to-safe, direction-specific. The single invalid cell is `deploy_production` on Mistral, where the *fidelity gate fails* (P(deploy)=0.06 — Mistral refuses to commit a production deploy even when instructed), so there is no commit state to brake — a model-behavior gap, not a brake failure. **The brake locus is a depth-relative late property** (∼80–98% of depth), whose exact layer is model-specific: Llama brakes uniformly at L26 (81% depth); Mistral at L32 (80%); Qwen's hardest actions need the final L63 (98%).

model	valid domains	brake works (valid)	redirect→safe	brake locus (depth)
Qwen3.6-27B	6/6	all	100%	L55 / L63 (86–98%)
Llama-3.1-8B	6/6	all	100%	L26 (81%, uniform)
Mistral-Small-24B	5/6	all	100%	L32 (80%)

Table 2: The brake law across three architectures. 17/18 valid cells; the brake works in all 17. The one invalid cell (Mistral × `deploy`) is a fidelity failure — Mistral will not commit a prod deploy — not a brake failure.

6 Discussion: what is shown, and its honest scope

What generalizes. The brake of [1] is not specific to a reversible edit, to one action, or to one model: across six genuinely irreversible actions and three architectures, a single late-layer, task-matched safe-donor patch collapses the agent's commitment and *redirects it to a safe, reversible action*. The redirect property is what makes this safety-relevant rather than merely destructive: a brake that turned a transfer into incoherence, or into a different transfer, would be useless; ours turns it into "check the balance."

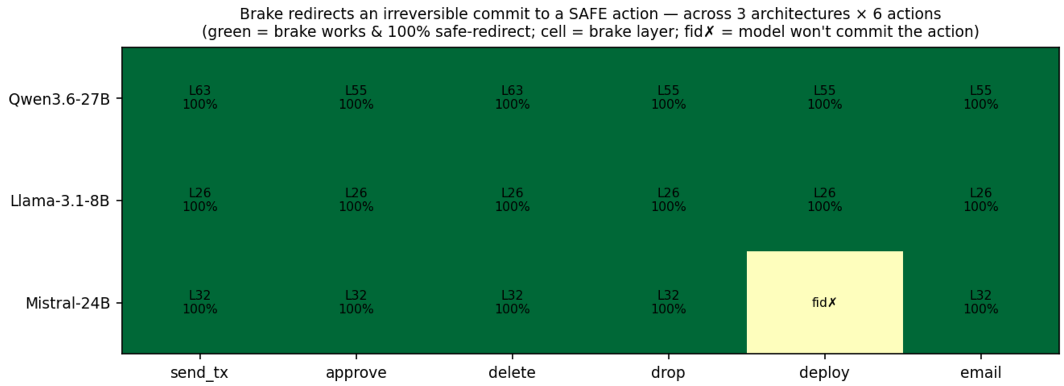

Figure 3: The brake redirects an irreversible commit to a safe action across 3 architectures × 6 actions. Green = brake works and redirects 100% to safe (cell shows the brake layer); `fid`× = the model will not commit that action, so there is nothing to brake.

What is not shown. (i) Actions are *simulated*; no funds move, no files are deleted. (ii) The intervention is a white-box, inference-time activation patch, so this is *mechanism generalization*, not a deployable defense. A single white-box direction is not robust to an adaptive adversary that can obfuscate activations [5], and an attacker who can patch activations already has access deeper than a same-host guard can defend; we therefore make no real-time-defense claim. (iii) The hardest actions (`send`, `delete`) are only *fully* braked at the *final* layer (`send` is 94% braked one layer earlier, but full suppression needs L63), leaving almost no margin for a guaranteed brake. (iv) Results are n=16–24 per cell, with a task-matched donor and one scenario family per domain; the "law" is across the tested actions and models, not proven universal.

Relation to prior work. This extends activation steering / representation engineering [6, 7] to an agent's *irreversible-action channel*. It is methodologically distinct from the training-time *Representation Rerouting* circuit breakers of Zou et al. [3], which fine-tune a model so harmful *content* representations become incoherent: we apply a single inference-time, late-layer patch at the agent's *action*-commitment point and, crucially, the agent does not become incoherent but *redirects to a safe action*. The brake direction is related to the refusal direction [4], but the steered quantity is the agent's commitment to an irreversible *action*, and the outcome is a redirect to a reversible alternative rather than a refusal. The result is consistent with mechanistic framings of *corrigibility* [8] — pausing or re-routing before an irreversible action.

7 Conclusion

The late action-commitment brake generalizes across six diverse irreversible agent actions and three model architectures, collapsing the commitment and redirecting the agent to a safe action with near-perfect reliability in this simulated, white-box setting. This establishes, at the mechanism level, that the "circuit-breaker" is a property of an agent's action channel broadly — not an artifact of one action or one model. The next steps are a clean threat model (the intervention is white-box and a single direction is not adversarially robust) and real, non-simulated actions.

Artifacts. The six-action battery, the cross-model harness, the per-action data, the figures, and an adversarial pre-publication evaluation are released at openinterp.org/research and in the public repository under `paper/circuit_breaker/` (`send_brake_test.py`, `multi_action_brake.py`, `multi_action_brake_xmodel.py`, ledgers `results/send_brake.json` and `results/multi_action_brake*.json`).

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